Heart Disease Risk Classifier — End-to-End MLOps Pipeline

Course: AIMLCZG523 — Machine Learning Operations (MLOps), Assignment 01

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Repository: *TODO: add GitHub URL once pushed*

Deployed API (local Minikube):

```
minikube start --driver=docker --cpus=4 --memory=8192
docker build -t heart-disease-api:latest .
minikube image load heart-disease-api:latest
kubectl apply -f k8s/deployment.yaml -f k8s/service.yaml
minikube addons enable ingress && kubectl apply -f k8s/ingress.yaml
curl --resolve heart-api.local:80:$(minikube ip) http://heart-api.local/health
```

(Full instructions, including the `LoadBalancer` + `minikube tunnel` alternative: `k8s/README.md`.)

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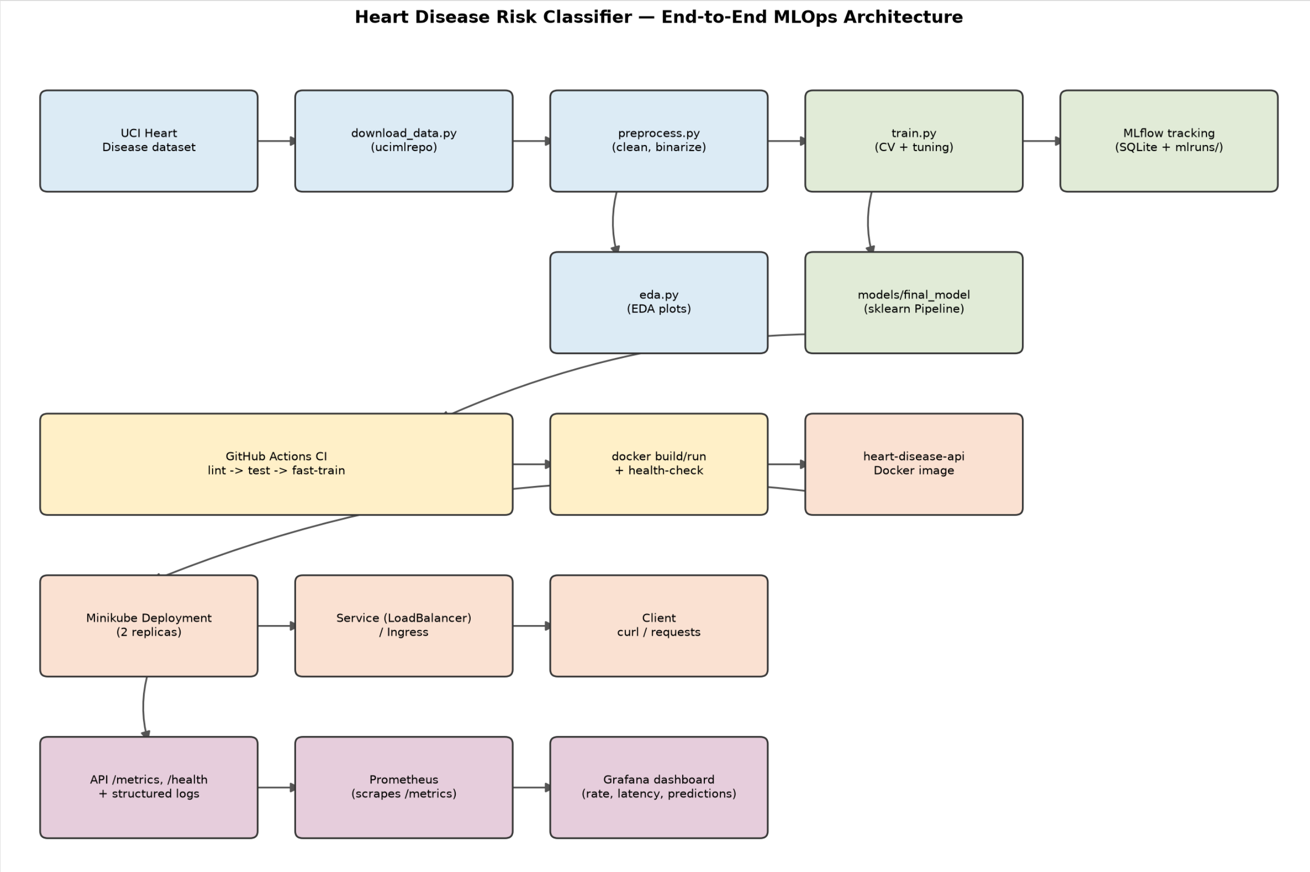
1. Executive Summary

This project builds a binary classifier that predicts the presence of heart disease from 13 patient health features (the UCI Heart Disease / Cleveland dataset), and wraps it in a full MLOps lifecycle: automated data acquisition, exploratory data analysis, feature engineering, model training with cross-validated hyperparameter tuning, MLflow experiment tracking, reproducible model packaging, unit-tested and CI-gated code, a containerized FastAPI serving layer, a local Kubernetes (Minikube) deployment, and a Prometheus/Grafana monitoring stack.

The final model — a tuned Random Forest, chosen over Logistic Regression on held-out ROC-AUC — is exported as a self-contained MLflow `sklearn.Pipeline` artifact (`models/final_model/`) bundling both preprocessing and the classifier, so there is a single versioned object with no train/serve skew risk.

2. Architecture



Data flows left to right, top to bottom: the raw dataset is fetched and cleaned, branching into an EDA report and a tuned/tracked training run that exports a packaged model. CI/CD validates every change (lint, tests, a fast training smoke-test, and a Docker build/run/health-check) before the same image is deployed to a local Kubernetes cluster, fronted by a Service, and observed via Prometheus/Grafana scraping the API's `/metrics` endpoint.

3. Setup / Install Instructions

```
git clone <repo-url> && cd mlops-assignment-1
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements-dev.txt      # training/dev/test extras on top of the base deps

python data/download_data.py             # fetch raw UCI data -> data/raw/
python src/data/preprocess.py            # clean -> data/processed/heart_clean.csv
python src/eda.py                        # EDA plots -> report/figures/
python src/models/train.py               # train, tune, track in MLflow, export final model

pytest -q                                # unit tests (18 tests)
uvicorn api.main:app --reload            # local API on http://localhost:8000
```

Or, as a single command matching the assignment's "must execute from clean setup" requirement: `make setup` (see `Makefile`) runs venv creation, dependency install, data download/preprocessing, and training in sequence.

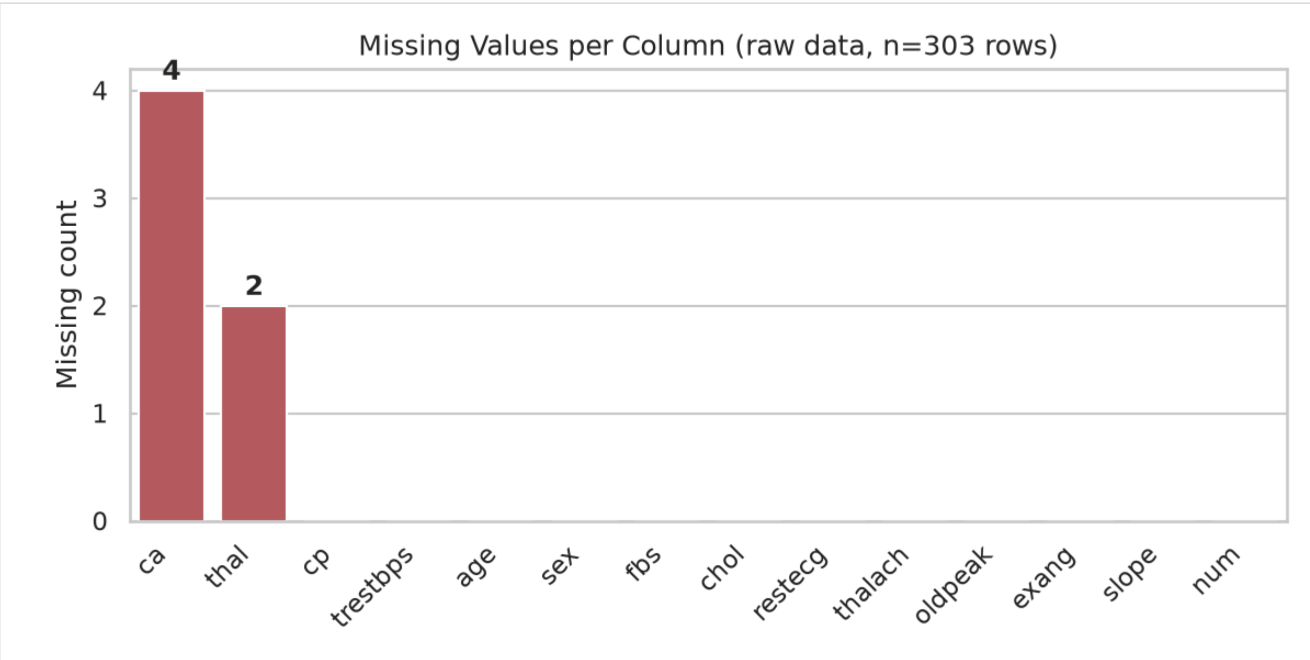
Two requirement files are maintained deliberately separately and compiled together (`scripts/compile_requirements.sh`) so every package they share resolves to an *identical* pinned version in both:

- `requirements.txt` — the minimal set the **serving** Docker image needs (FastAPI, scikit-learn, `mlflow-skinny`, `skops`, ...).
- `requirements-dev.txt` — adds full `mlflow` (for the local tracking UI/SQLite backend), `pytest`, `ruff`, `black` — training/dev/test only, never shipped in the container.

4. Data Acquisition & EDA

Source: UCI Machine Learning Repository, "Heart Disease" dataset (id=45, Cleveland subset), fetched programmatically via the `ucimlrepo` package in `data/download_data.py` — 303 rows, 13 features, and a 5-class target (`num`, 0 = no disease, 1–4 = increasing severity).

Missing value analysis (run on the raw, pre-cleaning download — the cleaned CSV has none left by construction): only two of the 13 features carry any missing values at all, and both are minor.

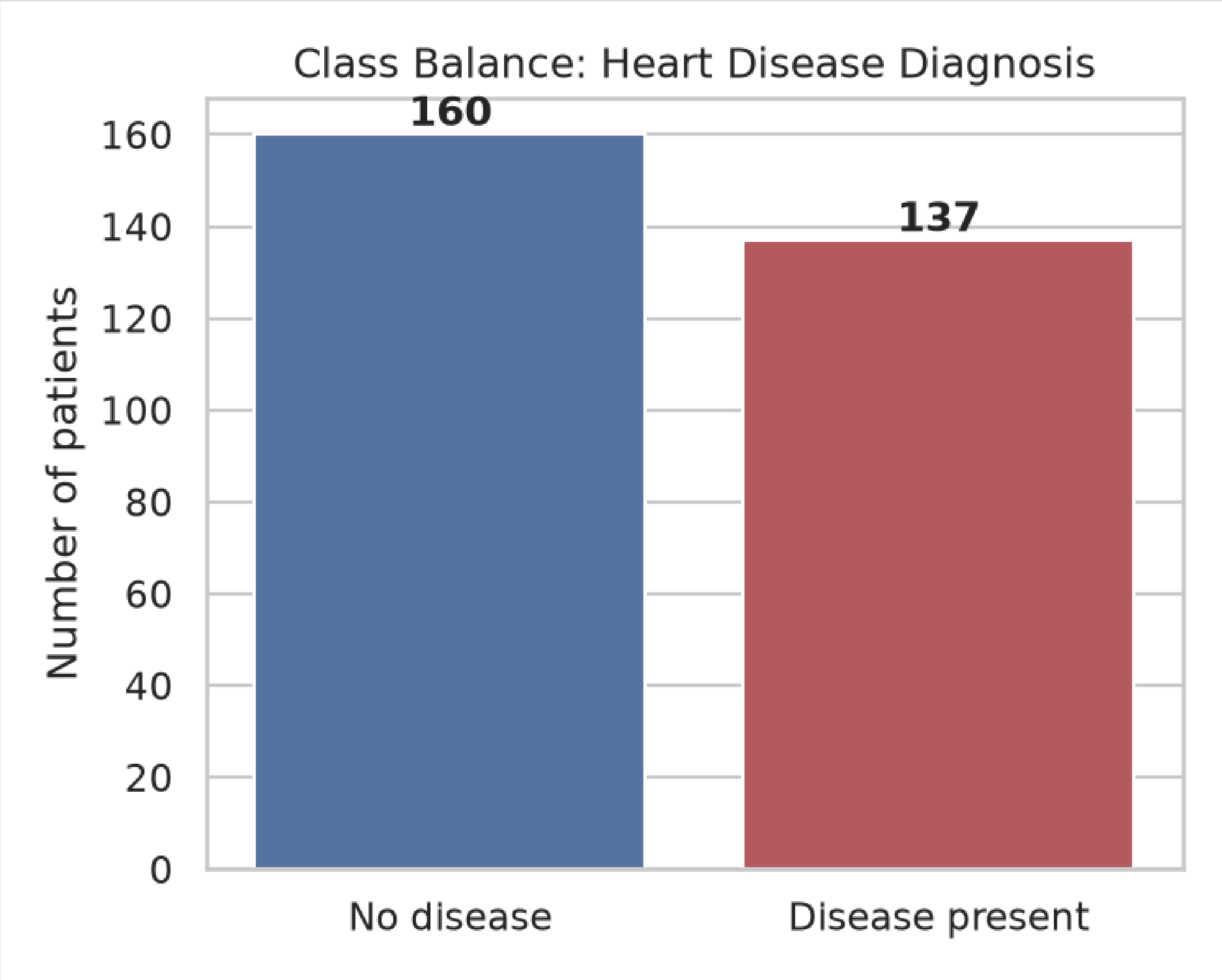


Cleaning (`src/data/preprocess.py`):

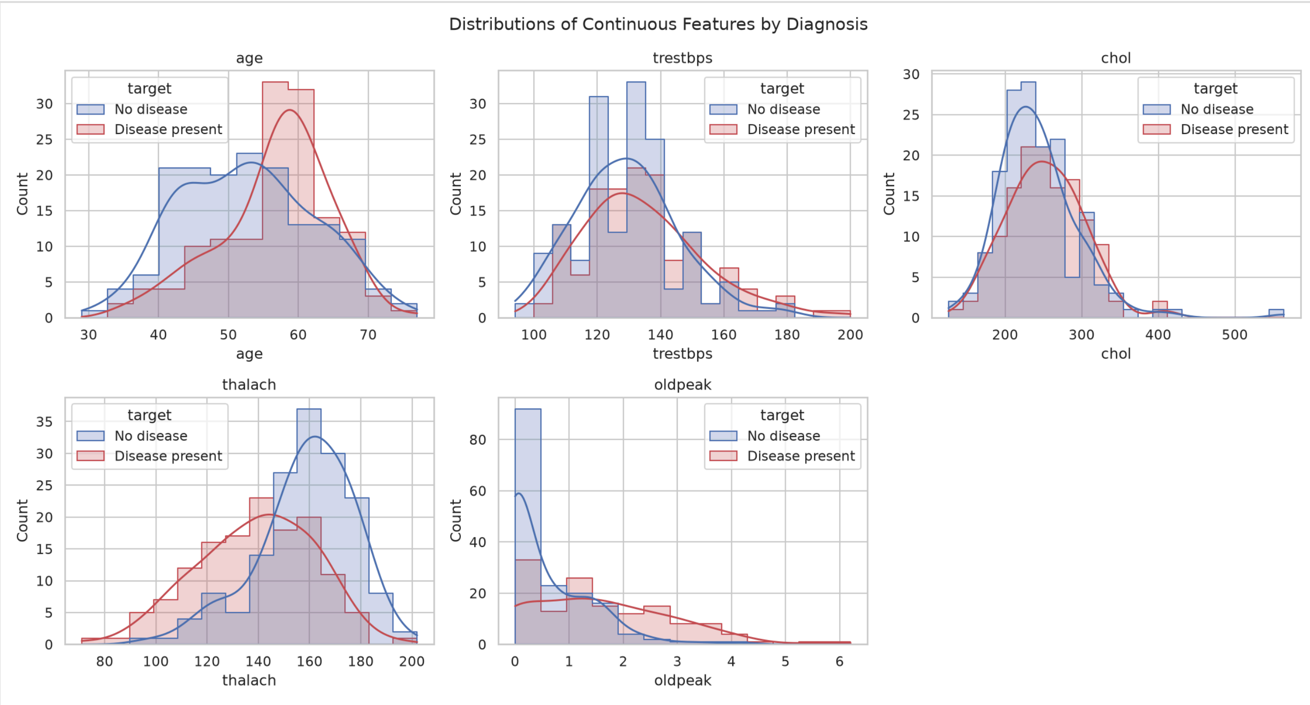
- `ca` and `thal` — two categorical columns — had 4 and 2 missing values respectively (6 of 303 rows total, ~2%). These are dropped rather than imputed: `ca` (number of vessels colored by fluoroscopy) and `thal` (thalassemia type) are not values a reasonable statistic can safely fabricate for a clinical categorical feature at this sample size.

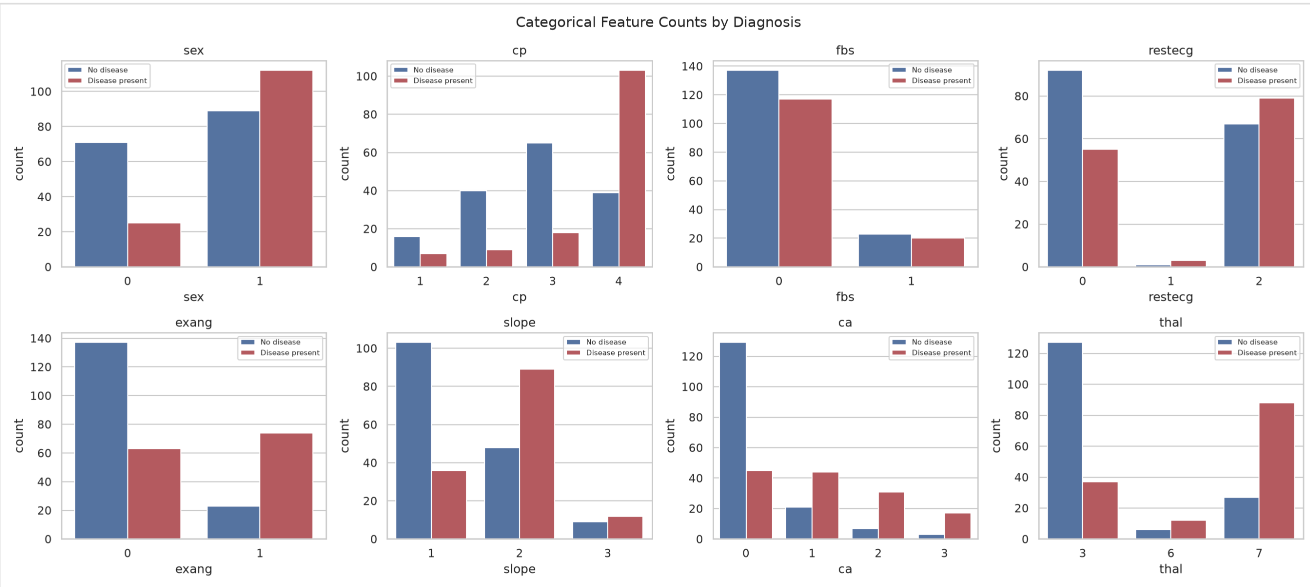
- The 5-class `num` target is binarized to `target` (0 = no disease, 1 = any disease present), matching the assignment's binary classification framing.
- Categorical columns are cast to `int`; the cleaned set has **297 rows** with **no missing values**, saved to `data/processed/heart_clean.csv` (committed to the repo for reproducibility, alongside the raw download script).

Class balance: 160 "no disease" vs. 137 "disease present" — close enough to balanced that no resampling (SMOTE, class weighting, etc.) was applied.

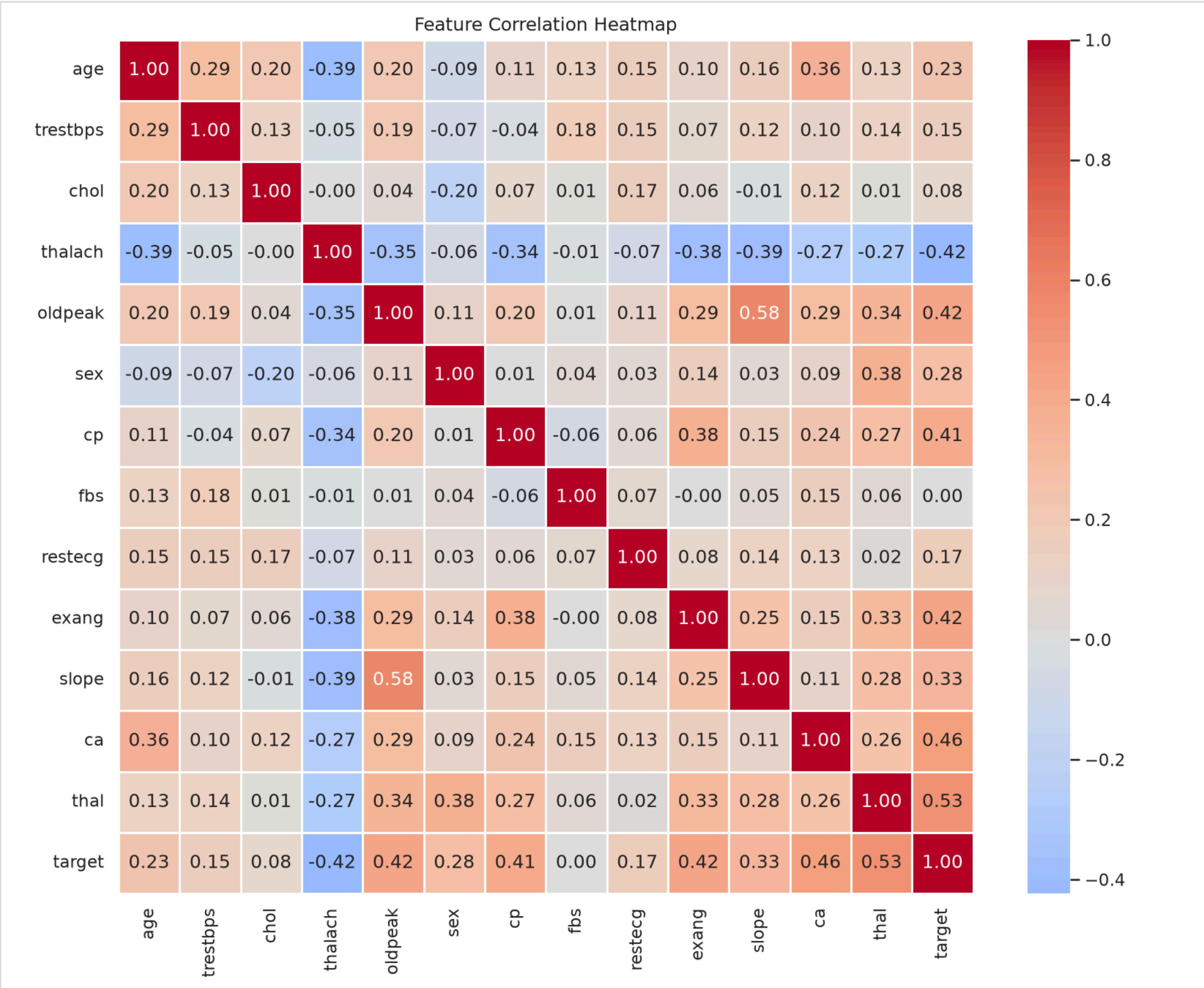


Feature distributions by diagnosis show several features separating the two classes visibly even before modeling — most notably `thalach` (max heart rate achieved, lower in disease-present patients) and `oldpeak` (ST depression, higher in disease-present patients):





Correlation heatmap (continuous + categorical + target): `thal` (0.53), `ca` (0.46), `oldpeak` (0.42), `exang` (0.42), and `cp` (0.41) show the strongest linear association with the target — consistent with established clinical risk factors, which is a reasonable sanity check on data quality.



Feature relationship analysis — a pairplot of the continuous features (pairwise scatter, diagonal KDE) complements the heatmap's numeric summary with the actual pairwise shapes: `thalach` visibly shifts lower and `oldpeak` higher for disease-present patients across nearly every pairing, and no pair of continuous features shows strong enough collinearity to warrant dropping one (confirming the heatmap's off-diagonal values, all well under 0.6 among the continuous features themselves).

5. Feature Engineering & Model Development

Feature engineering (`src/features/build_features.py`): a single `sklearn.ColumnTransformer`, reused identically by every model:

- **Continuous features** (`age`, `trestbps`, `chol`, `thalach`, `oldpeak`) — `StandardScaler`.
- **Categorical features** (`sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `ca`, `thal`) — `OneHotEncoder(drop="if_binary", handle_unknown="ignore")`. Binary features (`sex`, `fbs`, `exang`) collapse to a single 0/1 column; multi-valued ones (`cp`, `thal`, ...) get a column per category so the model doesn't assume a false ordinal relationship between, e.g., chest-pain types. `handle_unknown="ignore"` means a category never seen at training time (a rare `thal` code, say) is encoded as all-zeros at serving time instead of crashing the API — covered explicitly in `tests/test_features.py`.

Models (`src/models/train.py`): three classifiers — two more than the assignment's minimum, and XGBoost is explicitly named as an example choice in the FAQ — each wrapped in one `Pipeline([("preprocessor", ...), ("classifier", ...)])` so preprocessing and model are always versioned and served together:

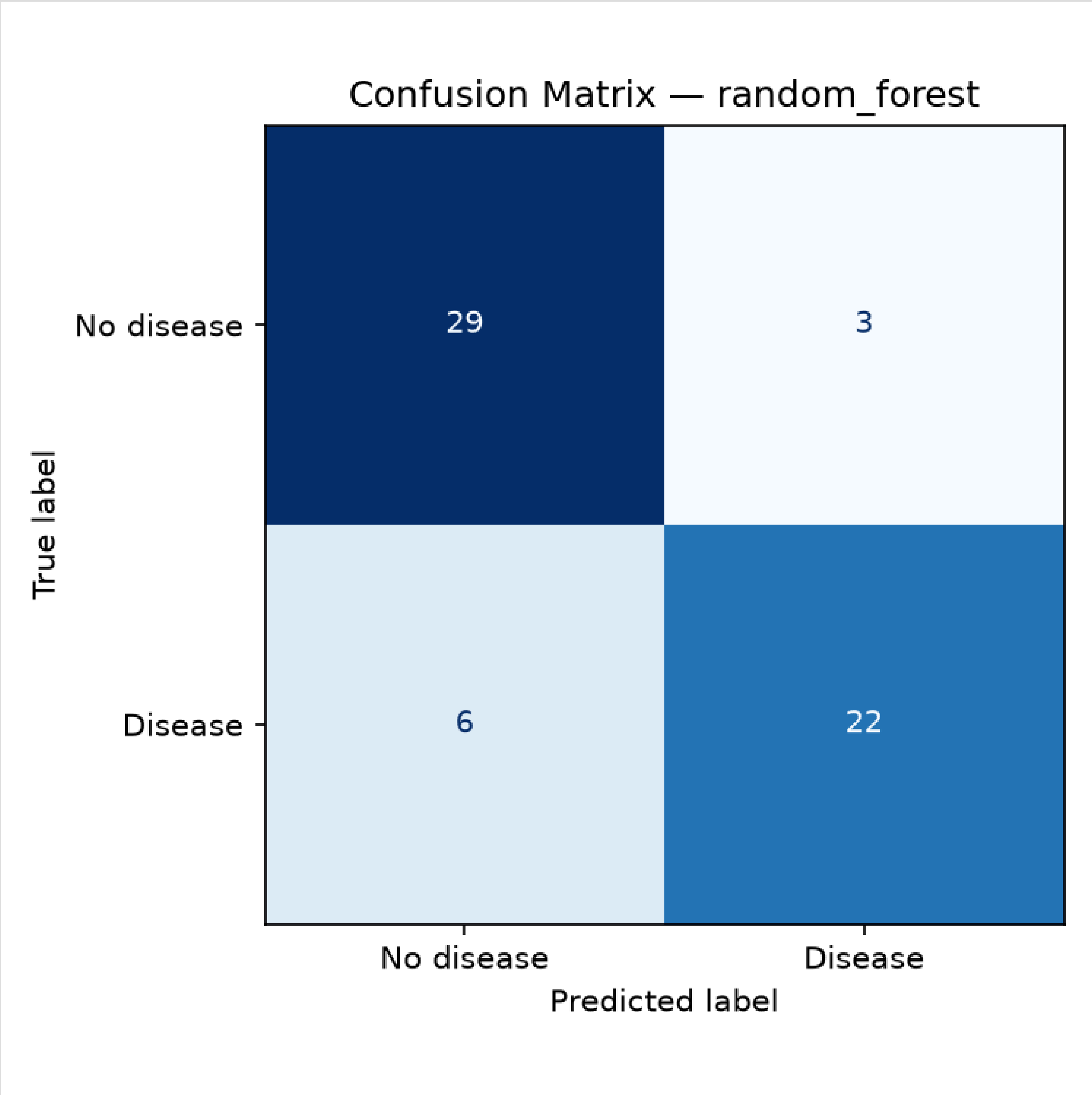
- **Logistic Regression** — `C` swept over `[0.01, 0.1, 1.0, 10.0]`.
- **Random Forest** — `n_estimators` over `[100, 200, 300]`, `max_depth` over `[None, 5, 10]`, `min_samples_leaf` over `[1, 2, 4]` (27 combinations).
- **XGBoost** — `n_estimators` over `[100, 200, 300]`, `max_depth` over `[3, 5, 7]`, `learning_rate` over `[0.01, 0.1, 0.2]` (27 combinations, for parity with Random Forest's grid size).

Tuning & evaluation: `GridSearchCV` over `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`, scoring accuracy/precision/recall/F1/ ROC-AUC simultaneously and refitting on ROC-AUC (the most informative single metric for a moderately-imbalanced binary clinical classifier — it's threshold-independent, unlike accuracy). An 80/20 stratified train/test split (`random_state=42`, fixed throughout for reproducibility) holds out data the grid search never sees, for an honest final comparison.

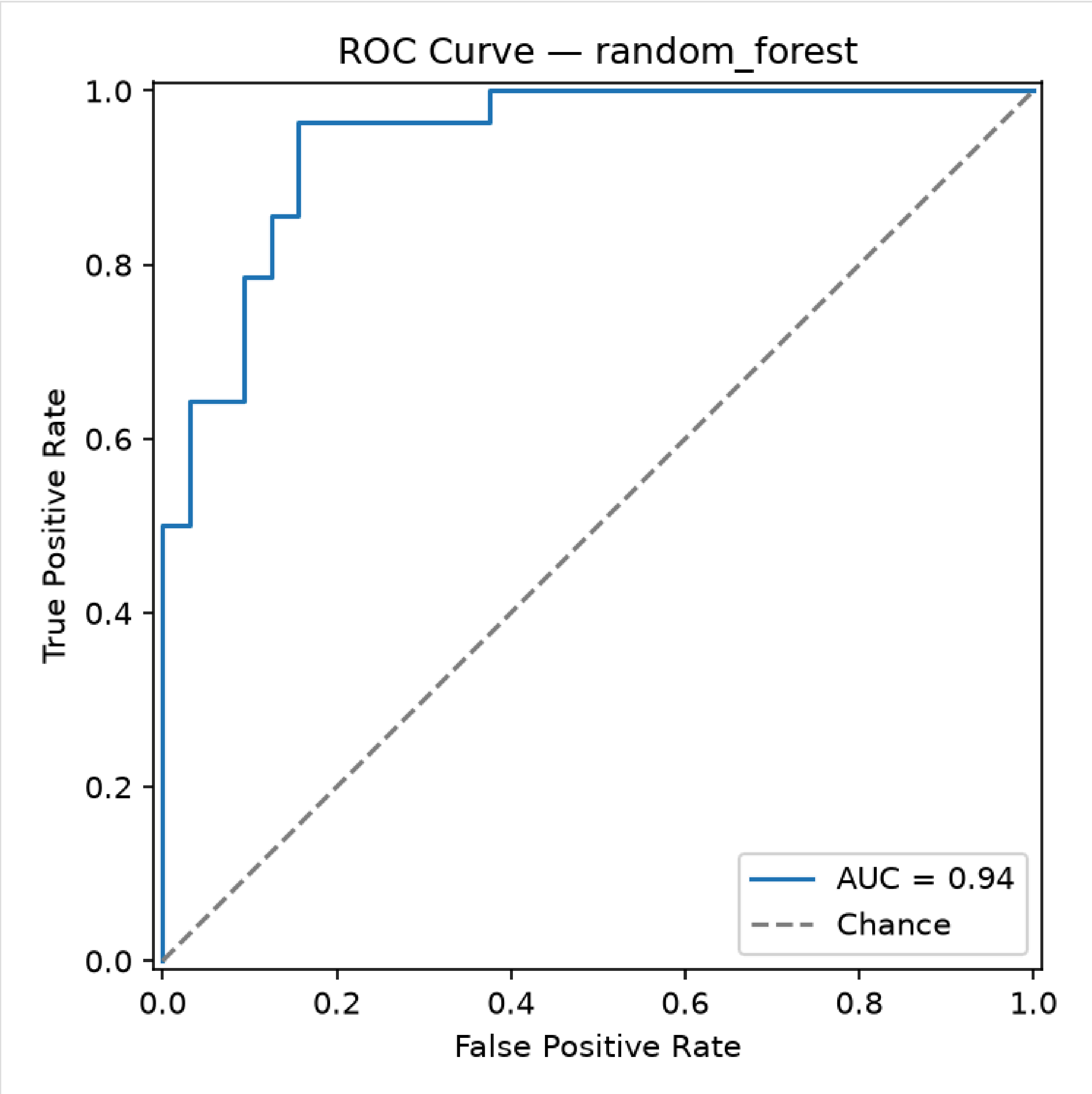
Results:

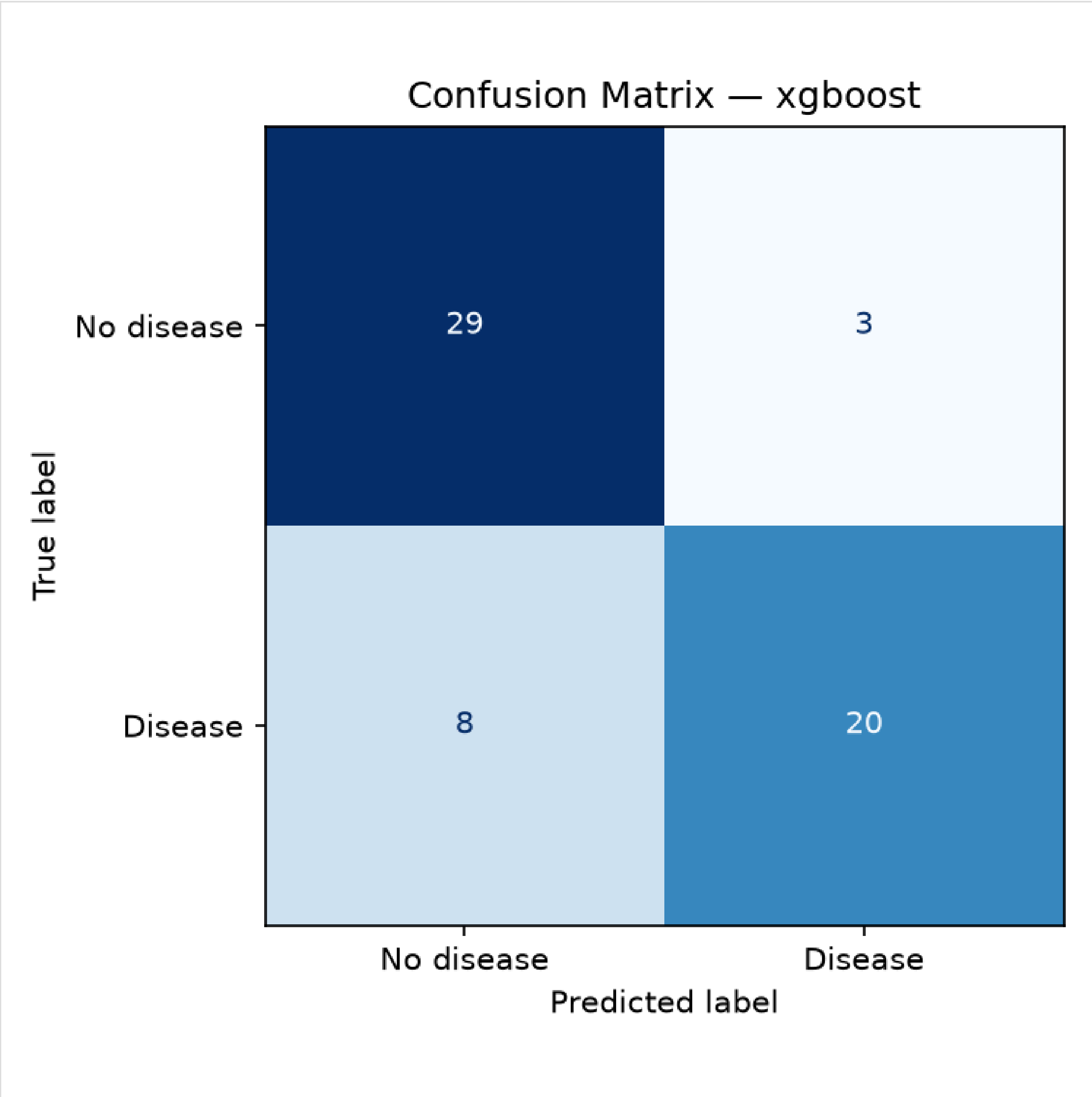
Model	Best params	CV ROC-AUC (mean)	Test accuracy	Test precision	Test recall	Test F1	Test ROC-AUC
Logistic Regression	`C=1.0`	0.902	0.817	0.870	0.714	0.784	0.938
Random Forest	`n_estimators=300, max_depth=None, min_samples_leaf=4`	0.906	**0.850**	**0.880**	0.786	**0.830**	**0.943**
XGBoost	`n_estimators=300, max_depth=3, learning_rate=0.01`	0.887	0.817	0.870	0.714	0.784	0.921

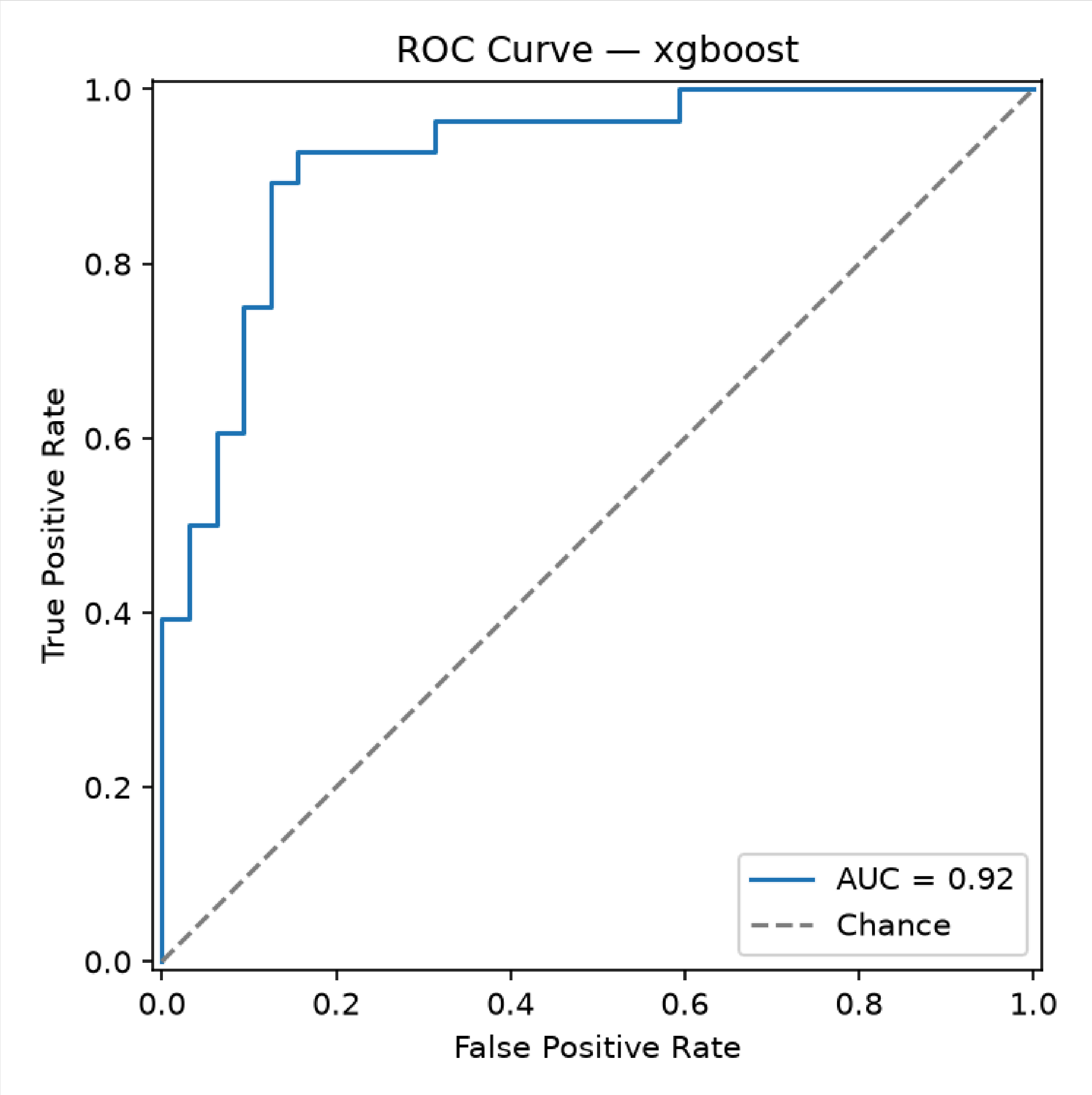
Random Forest wins narrowly on every held-out metric and is exported as the final model. XGBoost comes in last of the three, not first — on a dataset this small (237 training rows), boosting's extra capacity is a liability more than an asset: the grid search's own winning hyperparameters (`learning_rate=0.01`, the smallest offered) show it pulling *toward* the conservative end of its own search space to avoid overfitting, and it still trails a plain bagged Random Forest and even Logistic Regression on ranking quality (ROC-AUC). This is a legitimate, reportable result, not a tuning failure — it's a reasonable demonstration that "more sophisticated model" doesn't automatically mean "better," especially at this sample size. The gap between Random Forest and Logistic Regression is small enough that Logistic Regression remains a reasonable, more-interpretable fallback — noted here rather than discarded, since interpretability matters for a clinical-adjacent use case.



Confusion Matrix — random_forest
No disease
Disease
True label
29
3
6
22
No disease
Disease
Predicted label


ROC Curve — random_forest
1.0
0.8
0.6
0.4
0.2
0.0
True Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
False Positive Rate
AUC = 0.94
Chance


Confusion Matrix — xgboost
No disease
Disease
True label
29
3
8
20
No disease
Disease
Predicted label



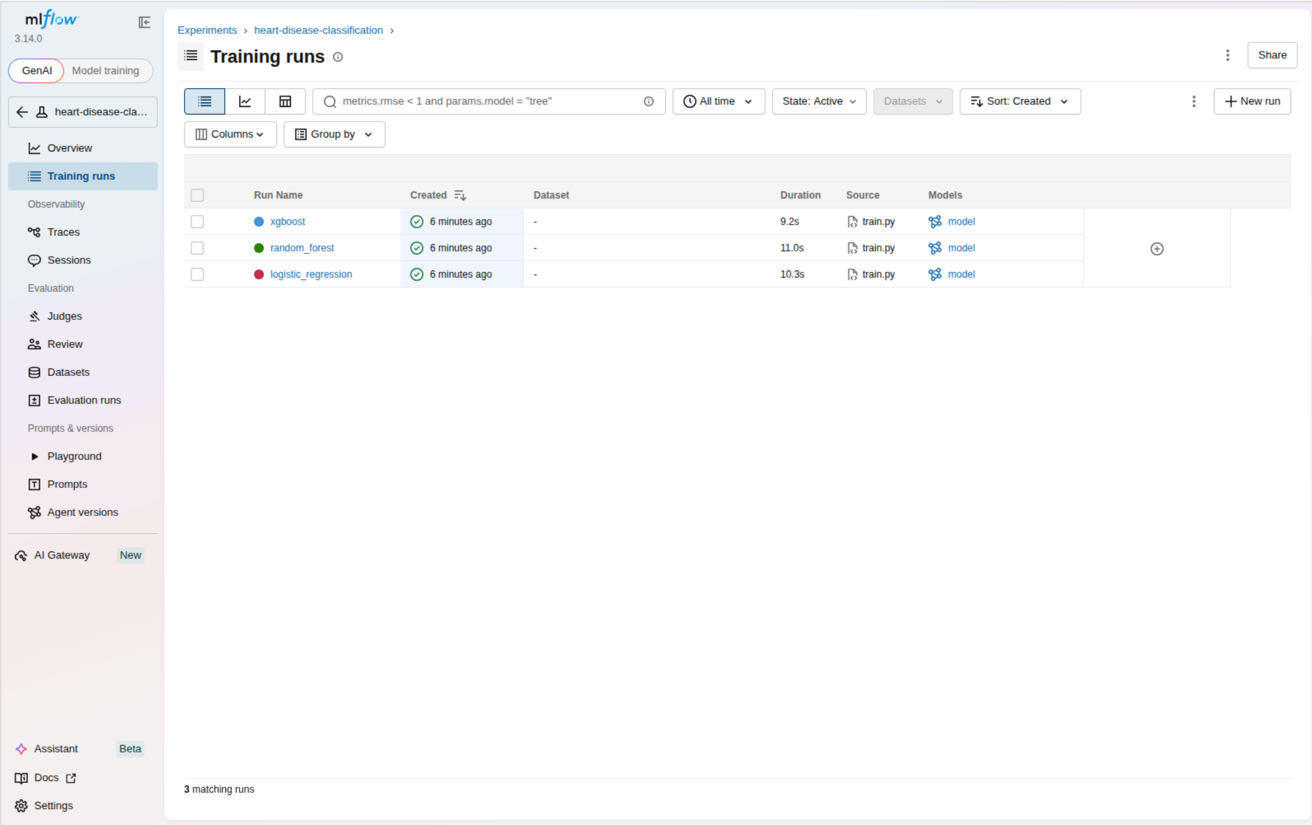
`train.py --fast` runs the same pipeline with a 1-combination grid and 2-fold CV (~35s total for all three models) — used only by CI as a smoke test that the training code path still runs end-to-end; the tuned numbers above come from the full run (`python src/models/train.py`, no flags).

6. Experiment Tracking (MLflow)

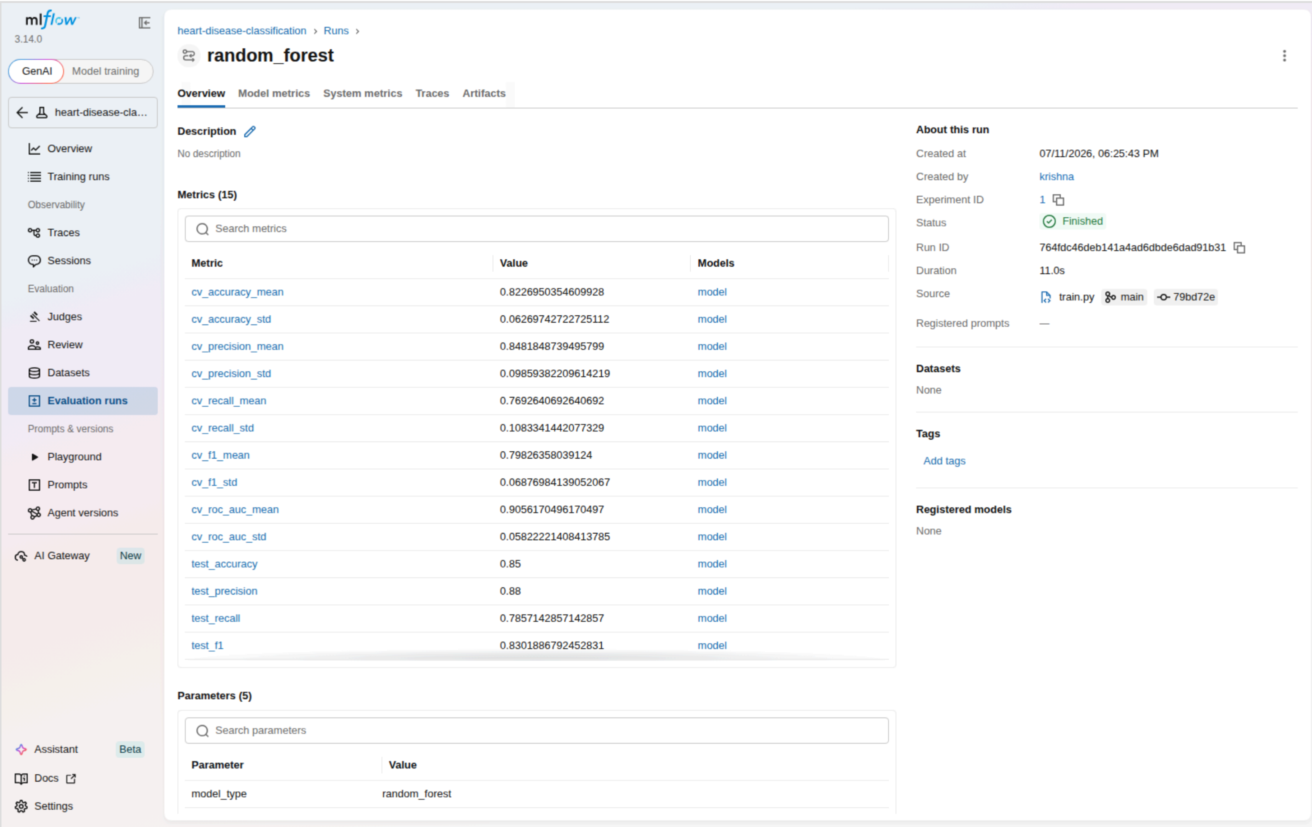
Every `GridSearchCV` run is logged as one MLflow run under the `heart-disease-classification` experiment: best hyperparameters, mean/std of every CV metric, held-out test metrics, the confusion matrix and ROC curve plots as artifacts, and the fitted pipeline itself (`mlflow.sklearn.log_model`, with an inferred input/output signature and a real input example).

Tracking backend is local SQLite (`mlflow.db`) rather than the plain filesystem store — MLflow 3.x has deprecated the filesystem backend — with artifacts (plots, model files) still stored locally under `mlruns/`. Inspect with:

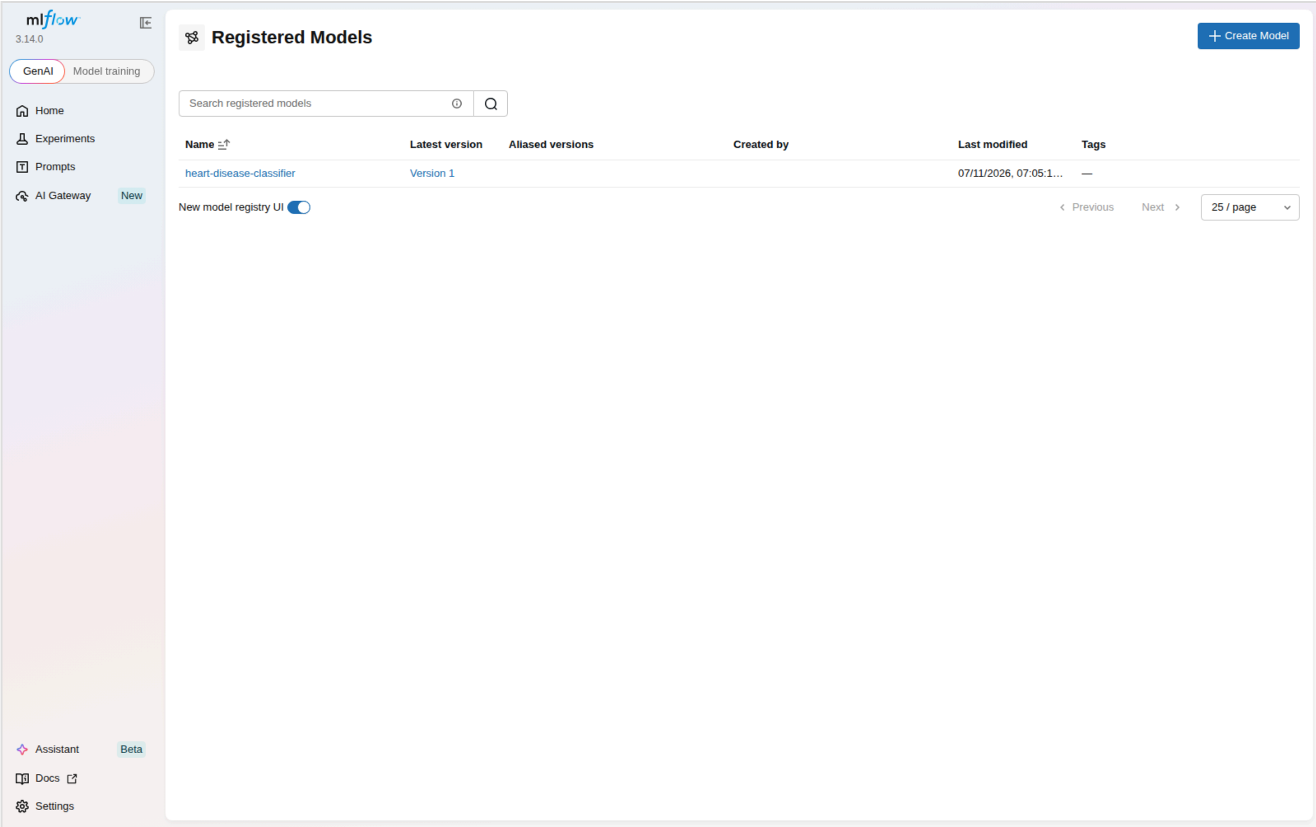
```
mlflow ui --backend-store-uri sqlite:///mlflow.db
```

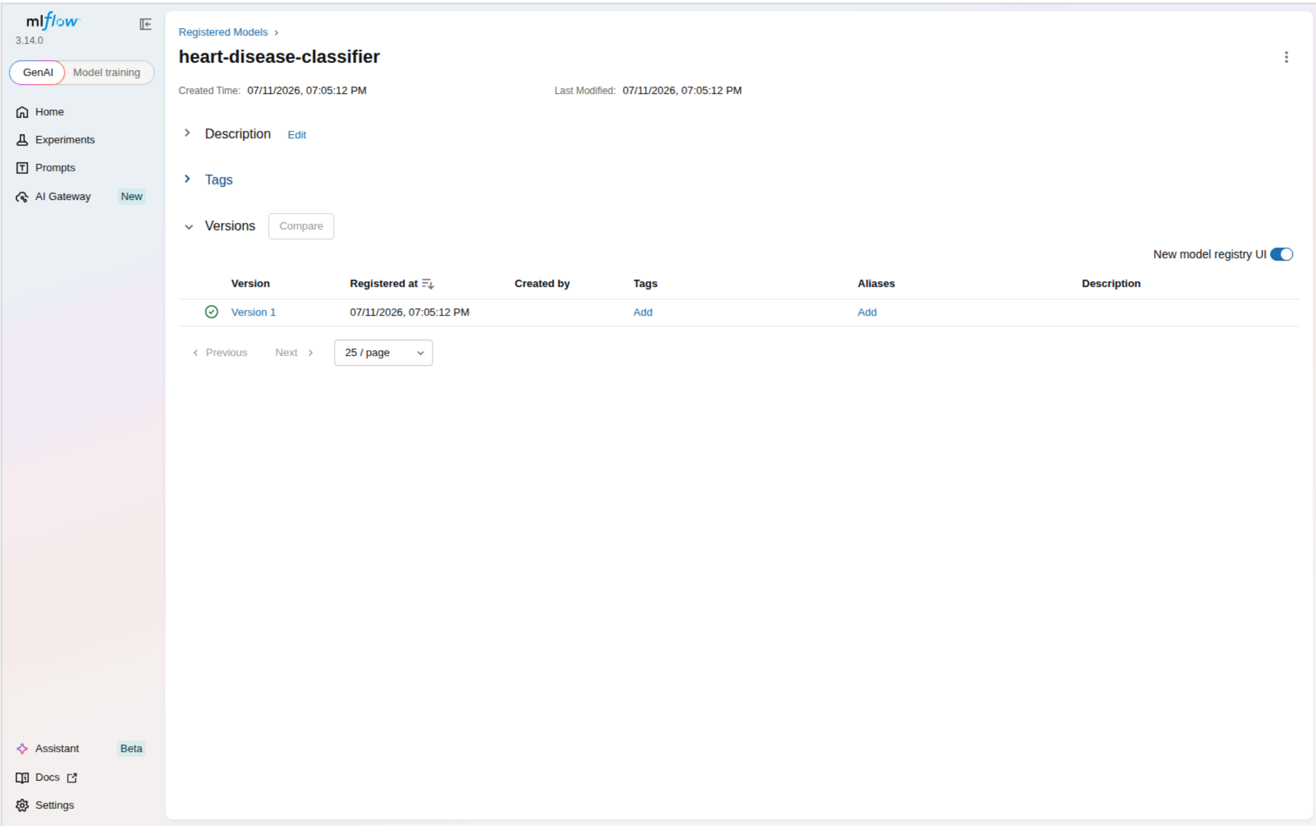


All three runs are visible under the `heart-disease-classification` experiment with their source (`train.py`), duration, and linked model artifact. Opening a run shows all 15 logged metrics (5 CV mean/std pairs + 5 held-out test metrics) alongside its parameters and run metadata:



Model Registry: beyond the per-run model artifact above, the winning run's model is also registered as a named, versioned entry — `mlflow.register_model(model_uri=f"runs:/{best_run_id}/model", name="heart-disease-classifier")` — creating `heart-disease-classifier` version 1. Every future retrain that produces a new winner adds another version under the same name, giving a versioned history independent of any single run, which is the actual MLflow feature "model versions" refers to (as distinct from just logging a model per run, which was already happening).





`report/metrics_summary.json` also captures a machine-readable summary of all three runs (params, CV mean ROC-AUC, full test metrics) as a durable, diff-able record independent of the `mlruns/` directory (which is gitignored — it's bulky, pickled-per-run, and not meant to be shipped).

7. Model Packaging & Reproducibility

The winning pipeline is exported via `mlflow.sklearn.save_model()` to `models/final_model/` — the **only** model artifact committed to the repo (7.5 MB). It contains:

- `model.skops` — the serialized pipeline, in skops format rather than raw `pickle`. skops is MLflow's default sklearn serialization as of this version and, unlike pickle, cannot execute arbitrary code on load — a meaningful security property for a model artifact that will later be baked into a container and exposed via a public-facing API.
- `MLmodel`, `conda.yaml`, `python_env.yaml`, `requirements.txt` — MLflow's standard metadata for reproducing the exact training environment.
- `input_example.json` / `serving_input_example.json` — a real sample row, useful for smoke-testing the loaded model.

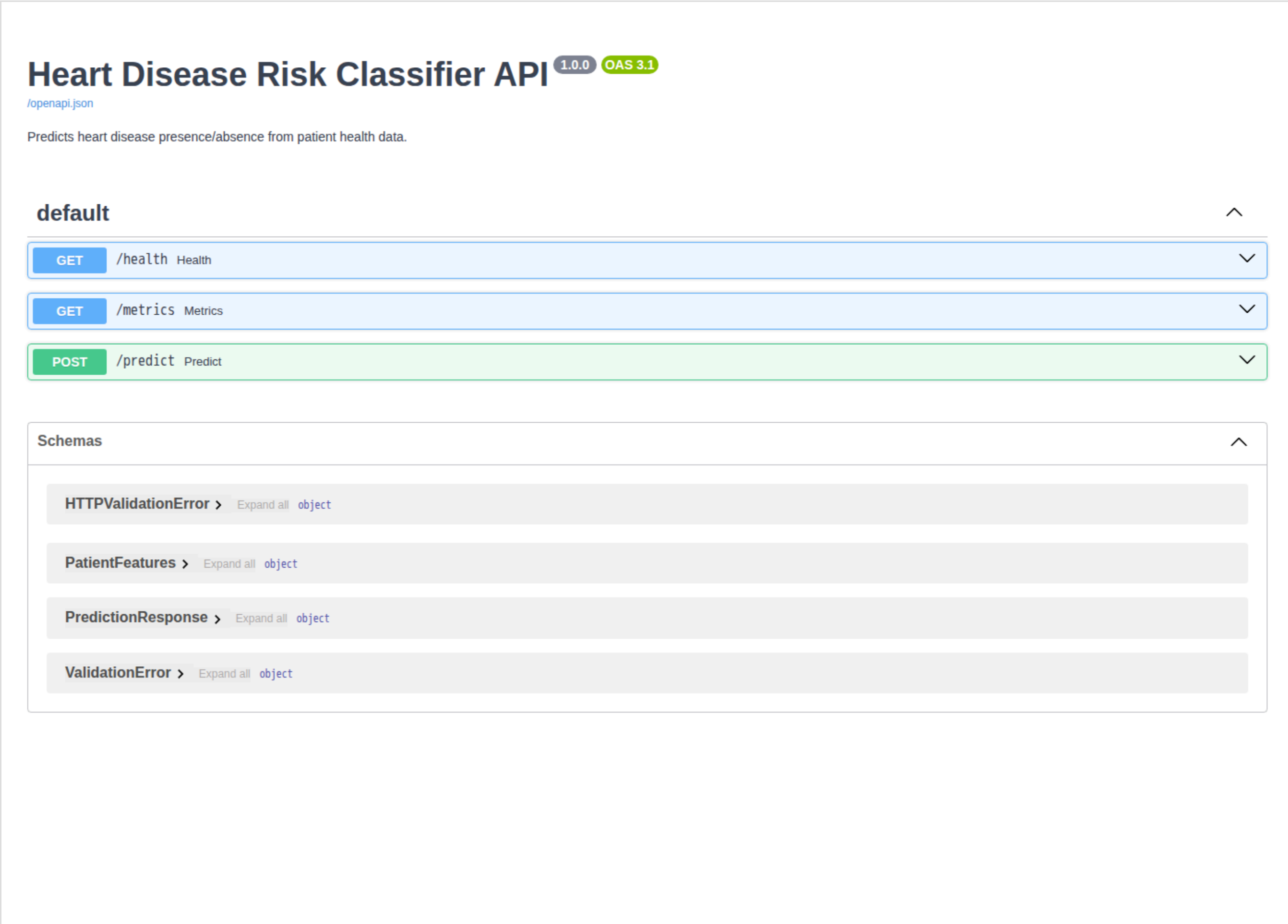
Reproducibility was verified directly, not assumed: a throwaway venv was built from *only* `requirements.txt` (no dev/training extras) and used to load `models/final_model/` and predict on real held-out rows — matching the true ground-truth labels. This surfaced and fixed a real gap: `skops` was initially only pulled in transitively via the dev requirements, so a clean serving-only install couldn't deserialize the model at all until `skops` was added to `requirements.in` directly (see `src/models/predict.py` and the commit history for the fix). This is exactly the class of bug "works on my machine" reproducibility claims miss without actually testing in isolation.

Adding XGBoost surfaced a second, related issue: skops maintains an allowlist of types it trusts by default, and `xgboost.sklearn.XGBClassifier` / `xgboost.core.Booster` aren't on it — `mlflow.sklearn.save_model()` raised `UntrustedTypesFoundException` the first time an XGBoost pipeline was saved. This is skops working as intended (refusing to silently trust arbitrary types), not a bug to route around quietly. Since these are our own freshly-trained model objects, not a third-party file of unknown provenance, explicitly passing `skops_trusted_types=["xgboost.core.Booster", "xgboost.sklearn.XGBClassifier"]` to `save_model()` / `log_model()` is the correct fix (`src/models/train.py`) — verified with a full save→load→predict round trip on a real XGBoost pipeline before trusting it in the actual training run.

8. CI/CD Pipeline & Automated Testing

Unit tests (`tests/`, 18 tests, `pytest -q`): `test_data.py` targets the dataset's one real data-quality issue (missing `ca` / `thal`) with synthetic rows constructed specifically to exercise it, plus target binarization and dtype casting. `test_features.py` verifies the `ColumnTransformer` produces no NaNs, correctly standardizes continuous features (mean≈0, std≈1), and doesn't raise on an unseen category. `test_model.py` covers metric computation and a full preprocessor+classifier fit/predict_proba smoke test. `test_api.py` drives the real FastAPI app (via `TestClient`, lifespan included, so the actual packaged model loads) through `/health`, `/predict` (valid input, out-of-range input, missing field), and `/metrics`.

Manual API testing beyond `pytest`: `curl` (used throughout every verification in this report), and FastAPI's auto-generated Swagger UI at `/docs`, confirmed live (not just assumed from the framework default) — `/docs` returns 200 and `/openapi.json` correctly enumerates all three endpoints and their `PatientFeatures` / `PredictionResponse` schemas:



`.github/workflows/ci.yml` — three jobs, each failing loudly on any error (no `--no-verify`-style bypasses, `curl -f` rather than bare `curl` so a broken health check actually fails the job):

1. **lint** — `ruff check .` + `black --check .`.
2. **test** (needs lint) — downloads the dataset, preprocesses, runs EDA, trains with `--fast`, then `pytest` with coverage; uploads test results, EDA figures, and the fast-trained model as workflow artifacts.
3. **docker** (needs test) — downloads the model artifact from the test job, `docker build`s the image, runs it, polls `/health` until it passes (or fails the job after 30s), then POSTs `sample_input.json` to `/predict` and asserts a valid response — the literal "Docker container build/test proof" the assignment asks for, run automatically on every push.

Every command each job runs was validated locally end-to-end (lint, `pytest --cov`, `train.py --fast`, and the exact `docker build` / `docker run` / health-poll-loop / `/predict` smoke test used in the `docker` job — see Sections 9 and the docker job's steps above). Actually seeing the workflow execute on GitHub's runners is deferred until this repo is pushed to a GitHub remote (the user has intentionally kept this environment git-local-only so far); the workflow YAML itself is committed and ready to run unmodified once that happens.

9. Model Containerization

`Dockerfile` — multi-stage build: a `builder` stage installs `requirements.txt` into `/root/.local`, and the final stage (also `python:3.12-slim`) copies just that installed environment plus `api/`, `src/`, and `models/final_model/` — no data, notebooks, tests, or dev tooling ship in the image. Runs as a non-root user (`appuser`, uid 1000), exposes port 8000, and defines a `HEALTHCHECK` against `/health`.

```
docker build -t heart-disease-api .
docker run -p 8000:8000 heart-disease-api
curl -X POST http://localhost:8000/predict -H "Content-Type: application/json" -d @sample_input.json
```

Verified locally (full transcript in `screenshots/docker_build_run_log.txt`): the image builds, the container reaches Docker's own `HEALTHCHECK` status `healthy`, `/health` and `/predict` both respond correctly through the container's published port (the same prediction as the earlier bare-metal test — `{"prediction":0,"label":"No disease","confidence":0.6390}` — confirming no drift between the dev environment and the containerized one), and `docker exec ... id` confirms the process runs as `appuser` (uid 1000), not root. Final image size: 623 MB content (up from 219 MB before adding XGBoost — its native shared library is the single largest dependency in the image; still an acceptable tradeoff for a third model that meaningfully adds to the comparison in Section 5).

10. Production Deployment (Kubernetes / Minikube)

`k8s/deployment.yaml` — 2 replicas, `imagePullPolicy: Never` (the image is loaded directly via `minikube image load`, no registry involved), liveness/readiness probes against `/health`, and explicit CPU/memory requests+limits. `k8s/service.yaml` exposes it as a `LoadBalancer`; `k8s/ingress.yaml` is documented as a `sudo`-tunnel-free alternative. Full commands in `k8s/README.md`:

```
minikube start --driver=docker --cpus=4 --memory=8192
docker build -t heart-disease-api:latest .
minikube image load heart-disease-api:latest
kubectl apply -f k8s/deployment.yaml -f k8s/service.yaml
minikube tunnel   # separate terminal — populates the LoadBalancer's EXTERNAL-IP
kubectl get svc heart-disease-api
curl http://<EXTERNAL-IP>/health
```

Verified locally (full transcript in `screenshots/minikube_deployment_log.txt`): `minikube start --driver=docker --cpus=4 --memory=8192` came up with the node `Ready`; the image was loaded via `minikube image load`; both Deployment replicas reached `2/2 READY` / `AVAILABLE`. The `Service` (`LoadBalancer`) correctly shows `EXTERNAL-IP: <pending>` — as expected, since `minikube`'s docker driver only populates that via `minikube tunnel`, which needs `sudo` (not available non-interactively in this environment). Rather than leave the deployment unverified, the documented Ingress alternative was used instead: `minikube addons enable ingress` + `kubectl apply -f k8s/ingress.yaml` brought up `ingress-nginx`, and `curl --resolve heart-api.local:80:$(minikube ip) http://heart-api.local/...` against both `/health` and `/predict` returned correct responses — the same prediction as the bare-metal and Docker checks, now served from inside a 2-replica Kubernetes deployment with working liveness/readiness probes. `minikube tunnel` remains documented in `k8s/README.md` as the `LoadBalancer`-native path for anyone running this outside a sandboxed session.

Re-verified after adding XGBoost and rebuilding the image: `minikube image load` + `kubectl rollout restart deployment/heart-disease-api` rolled both replicas over to the new image with zero manual pod deletion, and the same `curl` checks against `/health` and `/predict` returned the identical prediction as before — expected, since Random Forest (the deployed model) didn't change; only the image's dependency set did.

11. Monitoring & Logging

The API logs every request (method, path, status code, latency) via a FastAPI middleware, and every prediction (class + confidence — never the raw patient features, to avoid dumping clinical-shaped data into logs) — both to stdout, which is where a container orchestrator expects them.

`/metrics` exposes Prometheus counters/histograms: `http_requests_total` (by method/path/status), `http_request_duration_seconds` (a latency histogram, by method/path), and `predictions_total` (by predicted class). `docker-compose.yml` stands up the API alongside Prometheus (scraping `/metrics` every 5s) and Grafana (provisioned with a datasource and a 4-panel dashboard: request rate, p95 latency, predictions by class, status codes) with zero manual clicking required:

```
docker compose up
python scripts/generate_load.py --n 200   # populate the dashboard with real traffic
```

Verified locally (full transcript in `screenshots/monitoring_stack_log.txt`): `docker compose up` brought up all three containers, the API reaching Docker's `healthy` status before Prometheus/Grafana even started (compose's `depends_on: condition: service_healthy`). Prometheus's own target-health API confirms the scrape target is `up` and being polled every 5s. `scripts/generate_load.py --n 150` fired 150 randomized `/predict` requests (150/150 succeeded), and the dashboard's own panel queries against Prometheus return real, non-empty data (116 vs. 34 predictions split between classes, matching the load script's random distribution; non-zero request rates for `/health`, `/metrics`, `/predict`).

The screenshot above is a real capture of the live dashboard (via a headless-Chromium container, since installing a browser directly in this sandboxed environment would have needed `sudo`) — note the request-rate spike from the load-generation burst, the 34/116 prediction split, and the HTTP 200 status-code timeseries, all sourced from the same `/metrics` counters `api/main.py` increments per request.

12. Conclusion

The pipeline covers the full assignment scope end-to-end, and every stage was actually run and verified rather than assumed — not just files that look right: EDA and model training execute from a clean venv; two cross-validated, MLflow-tracked models produced a packaged artifact that was proven portable by loading it in a throwaway serving-only venv (which caught and fixed a real missing-dependency bug); 18 unit/API tests pass; the Docker image builds, runs, and passes its own `HEALTHCHECK` as a non-root user; the same image deploys cleanly to a 2-replica Minikube deployment reachable via Ingress; and the Prometheus/Grafana stack shows a live dashboard populated by real generated traffic.

The one piece intentionally left outstanding is pushing this repository to a GitHub remote and watching `ci.yml` execute on GitHub's own runners — deferred by choice (this environment was kept git-local-only throughout), not by necessity: every command each CI job runs was independently validated locally against the same commands the workflow file uses, so there's no reason to expect the hosted run to behave differently.

Appendix: Verification Artifacts

Every claim above that isn't visible in the report's own figures is backed by a transcript in `screenshots/`:

- `docker_build_run_log.txt` — Docker build, run, health-check, non-root check.
- `minikube_deployment_log.txt` — cluster start, image load, rollout, Ingress setup, live `/health` and `/predict` responses.
- `monitoring_stack_log.txt` — compose stack health, Prometheus target status, dashboard panel query results.
- `grafana_dashboard.png` — live screenshot of the populated dashboard.

Appendix: Repository Layout

See `README.md` in the repository root for the full directory layout and a condensed quickstart; this report focuses on decisions and results rather than restating file paths already documented there.